



11171 - Is the quest for Cosmic Dawn beyond all H0pe?

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				

Folder	Observation	Label	Observing Template	Science Target
	1	Plan_PLCKGz14+Lenses	NIRSpec MultiObject Spectroscopy	(1) Cycle5_MSA_Target+Lenses

ABSTRACT

The recent discovery of PLCKG165-z14-1, an ultra-blue and magnified $z = 14$ galaxy candidate behind the PLCK G165.7+67.0 cluster, presents an unprecedented opportunity to address fundamental questions within both Galaxy Evolution and Cosmology. At just ~ 300 Myr after the Big Bang, its exceptionally steep UV continuum slope ($\text{Beta} = -3.7 \pm 0.3$) implies dust-free stellar populations with a high escape fraction of ionizing photons and/or top-heavy IMF. Although intrinsically faint ($\text{MUV} = -18.7$), moderate gravitational lensing magnification ($\sim 3.3\times$) boosts PLCKG165-z14-1 to a brightness comparable to the "blue monster" galaxies ($\text{MUV} < -20$) at $z > 10$, making it uniquely accessible to JWST with a modest NIRSpec exposure time. With only 9 hours of NIRSpec PRISM MSA spectroscopy, we will obtain a definitive redshift via unambiguous detection of the Lyman break, while also achieving sufficient S/N to detect rest-frame UV emission lines. Any UV line detections will provide key insights into the ionizing photon production rate. This proposal will characterize the fainter galaxy population at $z > 13$, i.e. the galaxies which dominate the ionizing photon budget and are the primary driver of cosmic reionization. This multi-faceted proposal will also break new ground in Cosmology. The same MSA pointing will yield spectroscopic confirmation for at least seven strong-lensing systems in the G165 field, providing key lens model constraints. Combined with new time-delay measurements of the multiply imaged Supernova H0pe, these data will enable an independent, early-time measurement of the Hubble constant (H_0) to $< 5\%$ uncertainty, offering a powerful, novel approach to address the Hubble tension.

OBSERVING DESCRIPTION

We will observe the $z=13.9$ galaxy candidate PLCKG165-z14-1, situated in the PLCK G165.7+67.0 (G165) cluster field, using NIRSpec/PRISM MSA. Although PLCKG165-z14-1 is intrinsically low-luminosity ($\text{MUV} \sim -18.7$), the moderate magnification (~ 3.3) from the G165 cluster will allow us to perform a detailed spectroscopic analysis in only ~ 8.9 hours of observation. As well as providing unambiguous spectroscopic confirmation via detection of the Lyman break, this exposure time will also provide sufficient SNR/pixel to detect UV lines. At the same time, with the MSA we will observe a minimum of 7 lensed systems in the G165 cluster field, irrespective of the aperture position angle.

We will use the NRSIRS2 readout pattern, with 3 exposures (due to limits on exposure time), 16 groups per integration and 3 integrations per exposure. We will include a confirmation image with NRSIRS2 readout pattern and 5 conf. groups per integration. MSATA will be used for target acquisition. We will have a total of ~ 8.9 hours of science time, for which there will be ~ 12.3 hours of charged time.

Proposal 11171 - Targets - Is the quest for Cosmic Dawn beyond all H0pe?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Cycle5_MSA_Target+Lenses	RA: 11 27 14.3900 (171.8099583d) Dec: +42 28 29.04 (42.47473d) Equinox: J2000		
	Comments: Description=[]				

Proposal 11171 - Observation 1 - Is the quest for Cosmic Dawn beyond all H0pe?

Observation	Proposal 11171, Observation 1: Plan_PLCKGz14+Lenses										Sat Mar 14 00:00:28 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSspec MultiObject Spectroscopy										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	Cycle5_MSA_Target+Lenses	RA: 11 27 14.3900 (171.8099583d) Dec: +42 28 29.04 (42.47473d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		After Target ACQ and New MSA Config	MSA Center	z13+lens (54 sources)	fillers (61 sources)		jwst-nirspec-prism		1.5
Reference Stars											
Confirmation	#	Confirmation Type			Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations	Conf. Total Exposure Time	
	1	c1			NRSIRS2	5	1		1	379.311	

Proposal 11171 - Observation 1 - Is the quest for Cosmic Dawn beyond all H0pe?

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	171.78793870833 331 Degrees 42.471767222222 226 Degrees	41.985128926330 99			3	9	10635.301
	2	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	171.78793870833 331 Degrees 42.471767222222 226 Degrees	41.985128926330 99			3	9	10635.301
	3	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	171.78793870833 331 Degrees 42.471767222222 226 Degrees	41.985128926330 99			3	9	10635.301
Special Requirements	MSA Planned Aperture PA 42.0000 to 42.0000 Degrees (V3 263.4254303 to 263.4254303)										